Interacting Multipoles of Rank 2 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{array}{c}
+10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\epsilon,\zeta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta\left(\delta,\zeta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\zeta \cdot \delta\left(\delta,\epsilon\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\gamma,\zeta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\zeta \cdot \delta\left(\gamma,\epsilon\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot R_\zeta \cdot \delta\left(\gamma,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\beta,\zeta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \delta\left(\beta,\epsilon\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \delta\left(\beta,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \delta\left(\beta,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta\left(\alpha,\zeta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \delta\left(\alpha,\epsilon\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \delta\left(\alpha,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \delta\left(\alpha,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
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+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \delta\left(\beta,\zeta\right) \cdot \delta\left(\gamma,\epsilon\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\epsilon \cdot \delta\left(\beta,\gamma\right) \cdot \delta\left(\delta,\zeta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\epsilon \cdot \delta\left(\beta,\delta\right) \cdot \delta\left(\gamma,\zeta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
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+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\zeta \cdot \delta\left(\beta,\gamma\right) \cdot \delta\left(\delta,\epsilon\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\zeta \cdot \delta\left(\beta,\delta\right) \cdot \delta\left(\gamma,\epsilon\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
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+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\alpha,\beta\right) \cdot \delta\left(\epsilon,\zeta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\alpha,\epsilon\right) \cdot \delta\left(\beta,\zeta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
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\end{array} \right. \tag{1}$$

Simplify deltas:

$$
T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{array}{c}
+10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
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+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\zeta \cdot \delta\left(\beta,\beta\right) \cdot \delta\left(\delta,\delta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\delta\zeta} \\
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+105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\gamma,\gamma\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\
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+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\epsilon,\epsilon\right) \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\epsilon\epsilon} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\delta} \\
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+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \delta\left(\alpha,\alpha\right) \cdot \delta\left(\beta,\beta\right) \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\zeta}
\end{array} \right] \tag{2a}
$$

$$
\xRightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{array}{c}
+10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\delta\epsilon} \\
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-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\delta\zeta} \\
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-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\epsilon} \\
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-945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\delta\delta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\delta\delta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\epsilon\epsilon} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\delta\delta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\delta\delta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\epsilon\epsilon} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\delta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\delta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\delta\epsilon} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\epsilon} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\epsilon} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\delta\zeta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\zeta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\zeta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\delta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\delta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\epsilon\epsilon} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\delta\epsilon} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\delta\zeta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\zeta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\zeta} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\epsilon\epsilon} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\delta} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\delta} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\delta\epsilon} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\epsilon} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\epsilon} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\delta\zeta} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\zeta} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\zeta} \\
+105 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\epsilon}
\end{array} \right. \tag{2b}
$$

$$
\xRightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix}
+10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\epsilon} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\delta\epsilon} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\delta\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\delta\epsilon} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\delta\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\delta\epsilon} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\delta\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\epsilon} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\zeta} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\epsilon\zeta} \\
-945 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\
+210 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\delta\delta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\epsilon\epsilon} \\
+210 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\delta\delta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\epsilon\epsilon} \\
+210 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\delta} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\delta\epsilon} \\
+210 \cdot R_z^4 \cdot R_\alpha \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\epsilon} \\
+105 \cdot R_z^4 \cdot R_\alpha \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\delta\zeta} \\
+210 \cdot R_z^4 \cdot R_\alpha \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\zeta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\
+210 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\delta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\epsilon\epsilon} \\
+210 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\delta\epsilon} \\
+210 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\
+105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\delta\zeta} \\
+210 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\zeta} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\epsilon\epsilon} \\
+210 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\delta} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\delta\epsilon} \\
+210 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\epsilon} \\
+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\delta\zeta} \\
+210 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\zeta} \\
+105 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\epsilon} \\
+210 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\delta\epsilon} \\
+105 \cdot R_z^4 \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\zeta} \\
+210 \cdot R_z^4 \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\delta\zeta} \\
+105 \cdot R_z^4 \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\zeta} \\
+210 \cdot R_z^4 \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\zeta} \\
-15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} - 30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} \\
-30 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} - 60 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\
-90 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma}
\end{bmatrix} \tag{2c}
$$

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} +10395 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} \\ -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zzzz} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\delta\delta} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\delta\delta} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\delta\delta} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} \\ +210 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ -15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} - 30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} - 60 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\ 6 \qquad -90 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \end{bmatrix} \tag{3a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} - 945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} - 945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} - 945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -945 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} - 945 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} \\ -945 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} - 945 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} \\ -945 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} - 945 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} \\ -945 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} - 945 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} + 210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} \\ +105 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\epsilon\epsilon} + 210 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\delta\delta} \\ +105 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\epsilon\epsilon} + 210 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} + 210 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ +105 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} + 210 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ -15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} - 30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} - 60 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\ -90 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \end{bmatrix} \tag{3b}$$

$$\xRightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} - 1890 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} - 3780 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{zzz\beta} \\ -3780 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{zzz\alpha} + 105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\delta\delta} + 630 \cdot R_z^6 \cdot \Theta_{z\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +210 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} + 420 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{z\alpha} \cdot \Phi_{z\alpha\gamma\gamma} + 105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{zz\alpha\beta} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} - 15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} - 30 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} - 90 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \end{bmatrix} \tag{3c}$$

Simplify Quadrupoles:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} - 1890 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} - 3780 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\ -3780 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} + 105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} + 630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} + 420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} + 105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} - 15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} - 30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} - 90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{bmatrix} \tag{4a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} - 1890 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} + 105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} + 630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} + 420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} + 105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} - 15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} - 30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} - 90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{bmatrix} \tag{4b}$$

$$\xRightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -525 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} - 1050 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -1575 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} + 105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} + 105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} - 15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} - 30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} - 90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{bmatrix} \tag{4c}$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -525 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} - 1050 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -1575 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} + 105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} - 60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} \\ -90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{bmatrix} \tag{5}$$

Reduce Indices:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -525 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\alpha\alpha} - 1050 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\alpha\alpha} \\ -1575 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\alpha\alpha} + 105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} - 60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \\ -90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{bmatrix} \tag{6a}$$

$$\xRightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -3150 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\alpha\alpha} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} - 180 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{bmatrix} \tag{6b}$$

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -3150 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} - 3150 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} \\ -3150 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxxx} \\ +315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxyy} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\ +315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxyy} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyyy} \\ +315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\ +315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\ +1260 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxzz} + 1260 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} \\ +1260 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} - 180 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxxx} \\ -180 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxyy} - 180 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxzz} \\ -180 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxyy} - 180 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyyy} \\ -180 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} - 180 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\ -180 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} - 180 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \end{bmatrix} \tag{7a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot [+0] \tag{7b}$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = +0 \tag{8}$$